

Technical Configuration Reference Guide

Document Reference: CAW-TECH-CFG-2026-01
System: Case Ace v2.0 Client Consultation Recording & Note Drafting System
Organisation: Citizens Advice Wandsworth (CAW)
Status: Formally Approved Technical Reference
Classification: Internal / Technical Reference

1. Environment Variables Reference

ENVIRONMENT VARIABLES SPECIFICATION		
Variable Name	Default Value	Description
`NODE_ENV`	`production`	Application execution environment
`PORT`	`4000`	Backend HTTP server listening port
`HOST`	`127.0.0.1`	Local host bind address
`CORS_ORIGIN`	`https://caseace.adviceintelligence.tech`	Allowed client web origin
`GCP_PROJECT_ID`	`caw-case-ace-prod-2026`	Google Cloud Project identifier
`GCP_REGION`	`europe-west2`	Pinned UK sovereign GCP region
`GCP_SPEECH_RECOGNIZER`	`chirp2-uk-english`	Google STT v2 Recognizer resource
`GCP_VERTEX_MODEL`	`gemini-1.5-pro-002`	Large Language Model for drafting
`GCP_VERTEX_DRAFT_TEMPERATURE`	`0.2`	Low temperature for factual output
`SESSION_IDLE_TIMEOUT_SECONDS`	`900`	Inactivity timeout (15 minutes)
`AUDIT_LOG_RETENTION_DAYS`	`365`	Retention period for non-PII logs
`WEBEX_CLIENT_ID`	*(Encrypted Secret)*	Cisco Webex OAuth 2.0 Client ID
`WEBEX_REDIRECT_URI`	`https://caseace.adviceintelligence.tech/api/auth/webex/callback`	Webex OAuth callback URI
`ENTRA_REDIRECT_URI`	`https://caseace.adviceintelligence.tech/auth/callback`	Microsoft Entra ID callback URI

2. Client-Side Audio & Buffer Parameters

```
export const AUDIO_CONFIG = {
  sampleRate: 16000, // 16 kHz mono audio (optimal for Whisper & Chirp 2)
  channelCount: 1, // Mono recording channel
  maxDurationSeconds: 3600, // 60-minute hard session limit
  chunkSizeFrames: 4096, // Web Audio API ScriptProcessor / AudioWorklet buffer
  quantizationBits: 16, // 16-bit PCM Linear WAV
};
```

3. Local Whisper WASM Engine Parameters

```
export const WHISPER_CONFIG = {
  modelName: 'whisper-base-en-v3', // 74M parameter quantized model
  threads: 4, // WebAssembly SIMD worker threads
  temperature: 0.0, // Greedy decoding for maximum deterministic output
  language: 'en', // Enforce English language transcription
  beamSize: 5, // Beam search width
};
```

4. Prompt & Model Version Registry